Gwendal Beaumont

Ph.D. student in Software Engineering

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Education

2023 — Present **Ph.D. — Software Engineering**, *IMT Atlantique & IMT Mines Alès*, Brest, FR
Title of the thesis: TwinDevOps, an automated process for the deployment and dynamic evolution of Digital Twins.

2020 — 2023 **Engineer's Degree — General Engineering**, *IMT Mines Alès*, Alès, FR
One and a half year of general training and one and a half year in the Software Engineering and Artificial Intelligence Department.

2018 — 2020 **Classe préparatoire aux grandes écoles**, *Lycée Malherbe*, Caen, FR
Two-year intensive training to prepare for the French engineering schools competitive exam.
Main courses: Mathematics, Physics, Computer Sciences.

Experience

Jan 2026 — Present **Doctoral Visiting Researcher**, *TU Wien*, Vienna, AT
Doctoral mobility hosted by Assoc. Prof. Bork Dominik.
Detailed achievements:
- TBD: Ongoing mobility

Mar 2023 — Sep 2023 **Intern**, *ALTEN*, Rennes, FR
Research internship on the labelling process of data in order to be used with Machine Learning algorithms.
Detailed achievements:
- Creation of a framework using Java Spring Boot to manage unlabelled data and get statistics from it
- Creation of an interface using TypeScript and the Electron framework

May 2022 — Aug 2022 **Visiting International Scholar**, *Miami University*, Oxford, OH, USA
Internship as an assistant in the preparation of a master thesis on the effect of errors in agent-based simulation systems.
Detailed achievements:
- Creation of a simulation framework in Python to take errors into account
- Creation of multiple agent-based simulations in Python to test our framework

Leadership & Activities

2024 — 2025 **Ph.D. students representative at the Student Board**, *EULiST Alliance*
Institut Mines-Télécom Ph.D. students representative at the european level in the EULiST Alliance. This alliance integrates over 200,000 students and 20,000 staff at 10 universities in 10 countries.

2023 — 2025 **Ph.D. students representative at the board of directors**, *Institut Mines-Télécom*, FR
IMT Ph.D. students representative at the national level for two years. I take part in the board of directors and a work group focused on respect and equality.
- Creation of a national Ph.D. students network
- Creation of national events for Ph.D. students

2021 — 2022 **Communication Manager**, *IMT Mines Alès student council*, Alès, FR
Member of the communication managers team for the student council during a year.
- Administration of a website
- Management of social networks
- Answering and moderating emails and comments on social networks

Languages

French	Native	English	Fluent *(TOEIC: 975/990)*
Spanish	Intermediate	Korean	Basics

Computer skills

Coding	C/C++, Java, JavaScript/TypeScript, Python
Infra	Linux, Docker, Network

Articles

[1] Gwendal Beaumont et al. "Towards Automating the Life Cycle Management of Digital Twins". In: *Conceptual Modeling*. Ed. by Dominik Bork et al. Cham: Springer Nature Switzerland, 2026, pp. 412–430. ISBN: 978-3-032-08623-5. DOI: 10.1007/978-3-032-08623-5_22.

[2] Gwendal Beaumont et al. "Towards Re-Engineering Digital Twins: Preliminary Experiments on Three Use Cases". In: *Proceedings of the ACM/IEEE 27th International Conference on Model Driven Engineering Languages and Systems*. MODELS Companion '24. Linz, Austria: Association for Computing Machinery, 2024, pp. 453–458. ISBN: 9798400706226. DOI: 10.1145/3652620.3688259.

[3] Jack T. Beerman, Gwendal Beaumont, and Philippe J. Giabbanelli. "A framework for the comparison of errors in agent-based models using machine learning". In: *Journal of Computational Science* 72 (2023), pp. 102–119. ISSN: 1877-7503. DOI: 10.1016/j.jocs.2023.102119.

[4] Jack T. Beerman, Gwendal Beaumont, and Philippe J. Giabbanelli. "A Scoping Review of Three Dimensions for Long-Term COVID-19 Vaccination Models: Hybrid Immunity, Individual Drivers of Vaccinal Choice, and Human Errors". In: *Vaccines* 10.10 (2022). ISSN: 2076-393X. DOI: 10.3390/vaccines10101716.

[5] Jack T. Beerman, Gwendal Beaumont, and Philippe J. Giabbanelli. "On the Necessity of Human Decision-Making Errors to Explain Vaccination Rates for Covid-19: an Agent-Based Modeling Study". In: *2023 Annual Modeling and Simulation Conference (ANNSIM)*. Los Alamitos, CA, USA: IEEE Computer Society, May 2023, pp. 413–424. URL: https://ieeexplore.ieee.org/document/10155377.